

CS402 - Cloud & Distributed Systems

Scalable Service Deployment

Module code	CS402	Module weighting	25%
Deadline	Friday 16 October 2026, 17:00	Assessment type	Individual coursework

Overview

Design, implement and evaluate a small distributed application deployed using cloud infrastructure. The system must contain at least two independently deployable components communicating over a network interface.

Learning outcomes assessed

- Design a service-oriented architecture for a small distributed system.
- Implement secure communication between independently deployed components.
- Deploy, configure and observe the system in a cloud environment.
- Evaluate reliability, failure behaviour and likely scaling constraints.

Assessment criteria

1. Architecture and cloud design (20 marks)

- Define the service boundaries and responsibilities.
- Describe network communication and request/response contracts.
- Identify likely bottlenecks and single points of failure.
- Include a deployment diagram showing cloud resources and service relationships.

2. Service implementation (30 marks)

- Implement at least two independently deployable services/components.
- Expose and consume a documented API between components.
- Validate inputs and handle downstream or network failures.
- Keep configuration and secrets outside source code.

3. Cloud deployment (20 marks)

- Deploy the system to an appropriate cloud platform.
- Document environment configuration and deployment steps.
- Use repeatable build/deployment commands or automation.
- Demonstrate that a fresh deployment can be reproduced from the repository.

4. Reliability and performance (15 marks)

- Exercise normal and abnormal operating conditions.
- Demonstrate logging or monitoring sufficient to diagnose failures.
- Discuss scaling behaviour and at least two realistic bottlenecks.
- Provide one measured performance observation and interpret it cautiously.

5. Technical report (15 marks)

- Explain architecture and deployment decisions.
- Describe testing and failure scenarios.
- Discuss security, operational limitations and cost considerations.
- Maximum 2,000 words excluding references and appendices.

Submission requirements

- Source repository.
- Deployed application URL.
- Architecture/deployment diagram.
- Technical report, maximum 2,000 words.
- Short demonstration video, maximum 4 minutes.

Quality expectations

- Your submission should be internally consistent: requirements, implementation, tests and evaluation should refer to the same core behaviours.
- Where a feature is incomplete, describe the limitation accurately rather than presenting it as complete.
- Screenshots and demonstrations should show the actual submitted system.
- Evidence should be concise and relevant. Large volumes of generated documentation are not a substitute for working software or sound analysis.

Additional guidance

- The assessment focuses on distributed-system design and operational reasoning.
- A large microservice count does not automatically improve the submission.
- Any paid cloud resources should be kept minimal and documented.

Suggested evidence checklist

- Clear repository structure and meaningful commit history.
- Working primary user journey.
- Representative success and failure cases.
- Relevant automated tests plus manual end-to-end verification.
- Architecture/deployment explanation.
- Known limitations and future improvements.

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